DSA

Date :- July 06 2026

01. List of companies which are conducting placement drive through DSA.

01. Google
02. Amazon
03. Microsoft
04. Meta
05. Apple
06. Flip Kart
07. Adobe
08. Uber
09. salesforce
10. LinkedIn
11. Net flix
12. Twitter
13. Paytm

02. Syllabus of DSA in GATE

01. Introduction to Array Notes
02. Linked list notes
03. Binary Heaps notes
04. Heap sort Notes
05. Binary search Tree Notes
06. AVL Tree Notes
07. Graphs and their applications
08. Trees
09. stacks and their applications
10. Queues Notes
11. Introduction to Recursion

VARIABLES

01. Amazon.in Bill

```
String customer_name;
String order_id;
String product_name;
int quantity;
double gst;
double discount;
double grandTotal;
```

02. current Bill;

```
String customer_name;
String address;
String service_Number;
int units;
String bill_number;
String bill_date;
double tax;
double subsidy;
double total_amount;
String due_date;
```

03. Gas Bill

```
String customer_Name;
String address;
String mobile_number;
```

C

```c
#include <stdio.h>
&
struct student
{
int id;
char grade;
float cgpa;
};
int main()
{
int manual size=0, s.size
struct student s1; =0;
s.size = sizeof(struct
s1.id = 101; student));
s1.grade = 'A';
s1.CGPA = 9.5;
printf("ID: %d \n", s1.id);
printf("Grade: %c\n", s1.grade);
printf("CGPA: %.1f \n", s1.cgpa);
printf("SIZE: %d \n", s.size);
}
```

O/P:- ID: 101
Grade: A
CGPA: 9.5
SIZE:

CPP

```cpp
#include<iostream>
Using namespace std;
Struct student
{
int id;
char grade;
float cgpa;
};
int main()
{
student s1;
s1.id = 101;
s1.grade = 'A';
s1.cgpa = 9.5;
cout<<"Id"<<s1.id<<
cout<<"Grade:"<<s1.grad
<<e
cout<<"CGPA:"<<s1.cgpa
<<endl
return 0
}
```

O/P:- ID: 101
Grade: A
CGPA: 9.5

size (cpp)

```
#include<iostream>
using namespace std;
Struct Student
{
  int id;
  char grade;
  float cgpa;
}
int main()
{
cout<<"size :"<< sizeof(student);
  return 0;
}
```

op:-

size : 12

Manual:- int → 4
char → 1
float → 4
} 9 bytes

This is because padding (alignment)

H.W:-

Findout the applications of Recursion.

→ Is recursion still used in present programming era if so then where

→ what are languages that support recursive calls.

Function declaration :- Function prototype

Applicat^ns:-

01. Factorial
02. Fiboracci numbers
03. Binary search
04. Merge and Quick sort
05. Tree traversal
06. Graph Traversal
07. Back tracking pblms
08. Directory / File system traversal
09. parsing expression in compiler
10. Dynamic programming

→ used in present programming era

Yes

- Data structures
- AI
- compilers and parsers
- Operating systems
- Web developement

- competitive programming and DSA
- Game Developement

03. Languages support Recursion:

- C
- C++
- Java
- Python
- Java script
- C#
- GO
- Kotlin
- Swift
- Rust
- PHP

Date:- 10/07/2026

H.W:-

Find out the History of towers of Hanoi;;

Today from 7:30 AM, 4:30 PM

List out various Data structures you observed

HISTORY OF TOWERS OF HANOI

The Towers of Hanoi is a famous mathematical puzzle invented by the French Mathematician Edouard Lucas in 1883. He introduced it with legend about a temple where priest move 64 golden disks according to strict rules. The legend says that when the priests finishing moving all the disks, the world will end. Although the story is fictional it made the puzzle very popular. Today, The Towers of Hanoi is widely used to teach recursion, algorithm, and DS in CS.

- Move all 64 disks from first peg to the third peg.
- Move only one disk at a time
- Never place a larger disk on top of a smaller disk.

To solve that puzzle with 64 disk it would take

$2^{64}-1 = 18,446,744,073,709,551,615$ moves.

which is >>> 584 billion years with each second.

Morning :-

→ Alarm - sheeduling } Queue occur in order,

→ choosing clothes → Algorithm

→ packing your bag } stack last item packed is, of the first one taken out

eg :-

→ Attendance list → Array

→ searching for roll NO : → Linear search / Binary

→ waiting at canteen line → Queue

→ play list of the songs → linked list

Evng :-

→ social media friend Network → Graph

→ saving contact ; Hash Table

② i = 1 ; // 1
i = i+20 ; //i+1
pf("%d", i); // 1
} total : 4

How for all shell prgrms findout time-complexity in frequency count

③ for (i=1 ; i<n ; i++) ; → n+1
pf("Hello"); → 1
(n-1) → 1×(n-1)

= ';' indicates empty body loop which runs for 'n-1' times

∴ 1 + n/1 + n-1 + n-1+1

= 3n

③ i = 1; → ①
while (i<=n) → 1×n
{
pf("%d", i); → n
i++; → n
}

= 3n+1

H.W

Hash maps in java

Hash-maps in java — A collectn that stores data in key-value pairs.

ex:-

key	value
101 →	saifeela
102 →	Lohit

} value is accessed through key

Hashmap → dynamic in java

Creation of Hashmap:-

HashMap<Integer, String> student = new HashMap<>();

operations:- map.put("id", "value");

map.get("id");

map.remove("id");

map.size();

map.containsKey("id");

map.containsValue("id");

package → java.util.HashMap;

C → manually handles collision

ex:- index = key % size

java → key → hashCode() → Hash calculation

→ Bucket index → collision handling

→ storage → Resizing when necessary

Problems in society solved using DSA

		DSA soln	Helps with
01.	Emergency Ambulance routing	Graphs + shortest path	Find fastest route to a hospital
02.	Traffic congestion	Graphs + priority queues	efficient route and manage traffic flow
03.	Garbage collectn	Graphs + optimizatn	efficient routes for garbage trucks, reducing fuel / time
04.	Disaster Evacuatn	Graphs + BFS/DFS	Find reachable safe zones and evacuation routes
05.	Hospital Patient Management	priority Queue/ Heap	Emergency patients can be treated before less-urgent cases.